Hub suppression — stopping generic papers from hijacking niche reviews

arxivkg · generation · hub suppression

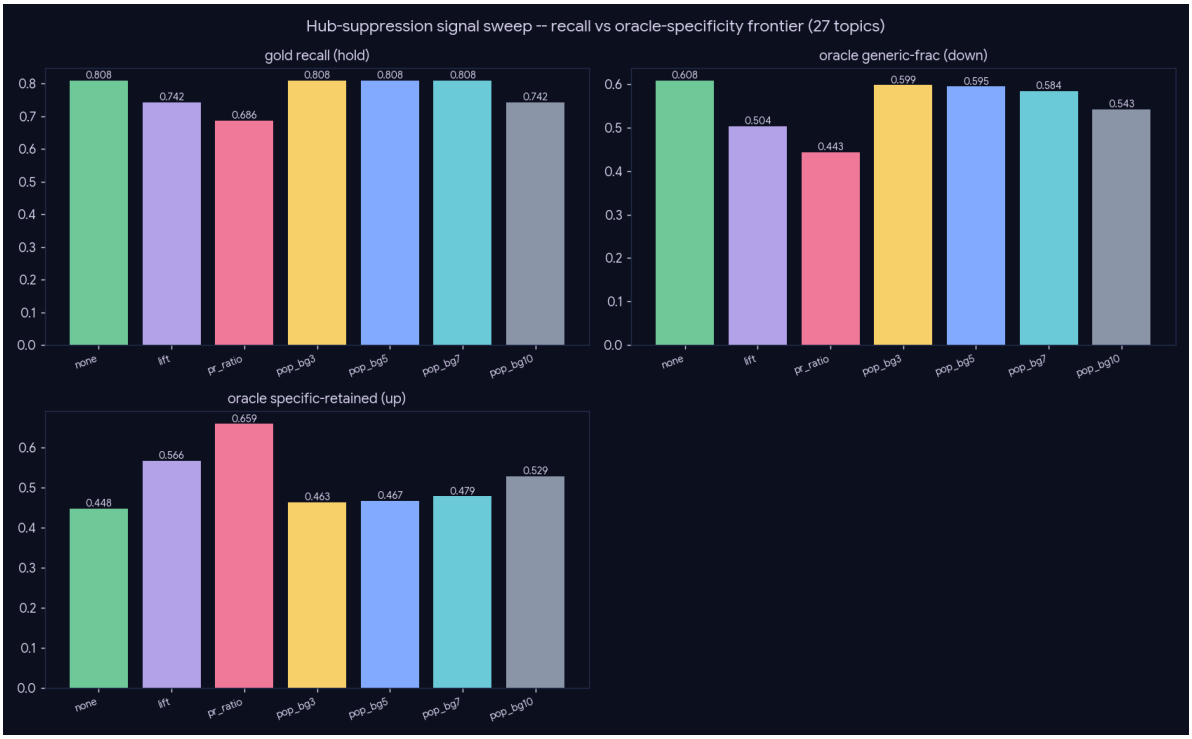
The problem

Reading the real deep-research outputs, the flagship's dominant rough edge is **generic-hub dominance**: "Attention Is All You Need" is cited in **27 of 27** deep reports, BERT in 17, GPT-3 in 16, the Adam optimizer in 11 — regardless of topic. A theorem-proving review devotes a whole "Rise of LLMs and Transformers" section to Attention→BERT→GPT-3 and cites *Adam as foundational to theorem proving*. Root cause: these hubs accrue high *in-topic* in-degree (nearly every modern paper cites them), so PPR/in-degree floats them to top landmarks. This is a **specificity** problem, not a similarity one — and it must be fixed without a blocklist.

Data-derived signal sweep (no blocklist)

We swept every plausible *data-derived* demotion signal — global IDF, topic-lift (P(cited|topic)/P(cited)), citation-intent background-ratio, a personalized-PR/global-popularity penalty, and `pop_bg` (demote only papers that are BOTH globally popular AND background-cited *here*, topic-conditioned, strength s). Two metrics over 27 topics: **gold_recall** (canonical retention — the tripwire) and an **independent LLM oracle** that labels each paper specific/generic *for that topic* (it never sees our signals).

arm	gold rec.	generic	specific ret.	bg
none	0.808	0.608	0.448	0.574
lift	0.742	0.504	0.566	0.598
pr_ratio	0.686	0.443	0.659	0.579
pop_bg3	0.808	0.599	0.463	0.565
pop_bg5	0.808	0.595	0.467	0.556
pop_bg7	0.808	0.584	0.479	0.545
pop_bg10	0.742	0.543	0.529	0.521



Finding: a real frontier, and no structural silver bullet

There is a genuine recall↔specificity **frontier**. The topic-conditioned `pop_bg` **holds gold recall perfectly** (0.808 → 0.808 at $s \leq 0.7$) because it never penalizes a load-bearing paper by fame alone — but it is *gentle* (oracle generic-frac only 0.608 → 0.584). Topic-blind penalties (`pr_ratio`) move the oracle most (0.608 → 0.443) but cost recall (0.808 → 0.686, concentrated on `transformer`, where the hubs genuinely *are* canonical). **Crucially, no purely-structural signal removes a hub that is genuinely *used* in the niche** — Attention stays a landmark in theorem-proving at every strength, because ATP papers really do use transformers (low background-ratio there). So "generic for this topic" is partly a **narrative** judgment, not a structural one.

The fix: a compose-side guard (shipped, A/B-validated)

Since retrieval cannot cleanly remove a legitimately-present-but-generic paper, the fix is a one-line compose guard (`compose_report._SYS` + `long_compose._OUTLINE_SYS`) forbidding a generic-foundations section unless directly load-bearing. **Deep A/B on "RL for automated theorem proving":**

	OLD (pre-guard)	NEW (guard)
"Rise of Transformers" section	present	**gone**
Attention / BERT / GPT-3 / Adam mentions	3 / 1 / 3 / 1	**0 / 0 / 0 / 0**
ATP-specific coverage (HTPS, DSP, Lean-STaR, ...)	present	**preserved**

Re-validated on three further niche topics (firm-network causal propagation; two document-layout-parsing variants): **zero** generic-hub mentions, all sections topic-specific, coverage intact — not

over-corrected. **The recall-safe `pop_bg` retrieval lever remains available** (behind a knob) for the milder generic-*method* residue in ultra-niche topics.

Caveat

The oracle is an LLM instrument (validated style of `03-01`); gold recall is n= 3. `pop_bg`'s structural metrics move modestly because it is deliberately gentle — its value is zero-recall-cost cleanup of truly-background hubs, complementary to the compose guard.